

At Niral Networks, we have done some indigenous end-to-end integration with a radio company in India, so that we can provide an end-to-end 5G system. Here's a description of our open and disaggregated network operating system called NiralOS.

NiralOS network operating system (NOS)

- **Private 5G core cloud:** Native private 5G core software for mobility, authentication, security, session and policy management. Niral 5G core also has a compact user plane function (UPF) to provide local breakout within an enterprise when integrated with TSP's centralised 5G core.
- **Mobile edge cloud:** Kubernetes and virtualised edge cloud infrastructure to create a mobile edge cloud (MEC) within an enterprise with open APIs to host third-party applications like AR/VR, robotics, drones, AI/ML, video analytics for low latency and privacy.

- **Centralised controller:** Provides centralised management, orchestration, zero touch provisioning and monitoring of multiple private 5G networks and MEC at various sites. The controller can be hosted in the public cloud to centrally manage and monitor multiple private 5G networks.

NiralOS specifications

- Release16-compliant 5G core for private 5G deployment
- 5G network functions--UPF, AMF, SMF, AUSF, UDM, NRF
- Kubernetes based cloud-native network function
- DPDK+VPP based UPF acceleration-linear scaling per CPU core
- UPF local breakout feature for

easy integration to TSP's centralised 5G core

- Support of N1, N2, N3, N4 and service based interfaces for 5G SBA
 - 5G core deployed on COTS hardware of various form factors and integrated with 5G radio
 - Kubernetes and virtualised, cloud agnostic edge cloud to host third-party applications
 - Open APIs for integration of third-party application to Niral 5G core and edge platforms
 - Web based dashboard for subscriber provisioning configuration and management
- Niral Networks has also been awarded for innovation in 5G. We look forward to achieving more such feats. **END** 🐧

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The author is founder and CEO of Niral Networks. This article is put together from a tech talk he gave at Open Source Conference 2022, an event organised by Samsung R&D Institute India – Bangalore, and IEEE ComSoc, Bangalore Chapter.

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